

What are U.S. undergraduate students really doing with GenAI?

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Lollipops & Lemons.

Lollipops: What makes you hopeful? Lemons: What is unpleasantly challenging?

DIRECTIONS

1. Take a notecard from the envelope.
2. Write: name · role · institution · one lollipop or lemon (or both)
3. Pass it around your table.
4. Share what surfaced — we'll hear a few.



3 minutes

Lollipops & Lemons.

Lollipops: What makes you hopeful? Lemons: What is unpleasantly challenging?

DIRECTIONS

1. Turn to a neighbor.
2. Share: name · role · institution · one lollipop or lemon (or both).
3. We'll capture a few on the flipchart.



3 minutes

A note before we go further.

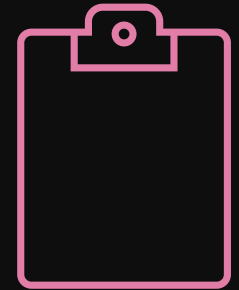
"You didn't ask for AI to arrive at our educational doorstep. . .

You didn't ask for an unregulated, untested, unreliable, untraceable, and unrelenting technology to upend education... it is clear the genie is out of the bottle... it is not going back in nor is it going to slow down for us to catch up" (Ludwig & Zakrajsek, 2026, p. 3).

Neither did your students.

01

What did the students tell us.



10,000+ surveys · 16 interviews · 2 years

GENERATIVE ARTIFICIAL INTELLIGENCE AND THE METALITERATE LEARNER: A MIXED METHODS STUDY OF AI-LITERACY AND US UNDERGRADUATE STUDENTS' INTERACTION WITHIN AN AI-INTEGRATED KNOWLEDGE ECONOMY (ALSTOT, 2026)

Study at a glance.

DESIGN Explanatory Sequential Mixed Methods	ANALYSIS Grounded Theory	SITE San Diego State Univ. Public R1 · HSI · 30,000+	SAMPLE 10,000+ surveys 16 interviews	INSTRUMENTS Survey Semi-structured interviews	TIMELINE Fall 2023 Fall 2024
RESEARCH QUESTIONS RQ1. What is AI literacy in a mid-21st-century GenAI-integrated knowledge network? RQ2. What is an AI-literate learner in a mid-21st-century GenAI-integrated knowledge network?			HYPOTHESES There is a relationship between AI use and: H1 awareness and understanding of AI H2 perceptions and attitudes toward AI H3 opportunities for students to learn about AI at their institution H4 students' future expectations of AI H5, H6 <i>student demographic variables — relationships too weak to determine</i> <i>For full explanation, see Chapter 4, pp. 207–228.</i>		

What the numbers tell us.

Highlights from the quantitative data

WHAT HELD

Four of six hypotheses supported.

H1, H2, H3, H4 showed positive relationships with AI use.

H3 *is the one to watch.*

WHAT SHIFTED · 2023 → 2024

Use went up. Almost everything else weakened.

- ▼ Awareness (H1)
- ▼ Attitudes (H2)
- ▼ Future expectations (H4)

▲ Desire to learn about AI (H3)

THE DISCONNECT

Substantive disconnect between high adoption and low engagement in AI education. (Fall 2024 Survey)

Students are using AI more — but not engaging in “opt-in” AI education.

What factors contribute to this disconnect between AI-use and AI-readiness?

I sat down with 16 undergraduate students.



First, what we heard.

F1

An AI-integrated learning process is values-driven.

*Shaped by **learner context** and **foundational AI knowledge**.*

"I don't want to use it to cheat. I want to use it to help understand what I'm studying... not something to basically do my work for me."

— Lyla, Y5+ · Geology

"I wouldn't have minded just doing the assignment, but it was the fact that I had to hit an arbitrary word count... there's no other way for me to get this done. So I used AI."

— Gabriel, Y3 · Marketing Comm

And second.

*F*₂

Co-regulated AI interactions drive **metaliterate** genAI integration.

Learning happens in the loop between learner and AI.

"I'll typically go to ChatGPT and specifically ask for it not to generate — which is what it thinks I want, which is to just write the essay for me. I explicitly tell it, do not write this for me. I need to do this on my own, but I want help with understanding what I should write about."

— Ian, Y1 · Computer Science

Meet the nine.

Each table will receive an envelope with one or two of their stories.

Amber

Y1 · Communications

career returner · age 61

Diana

Y2 · Child & Family Dev

first-gen · ESL

Gabriel

Y3 · Marketing Comm

first-gen

Harper

Y1 · Political Science

traditional · biracial

Ian

Y1 · Computer Science

processing disorder · Pell

Nicholas

Y1 Electrical Engineering·

first-gen · Pell

Lyla

Y5+ · Geology

ADHD · Pell

Miguel

Y1 · Computer Science

first-gen · Pell

Phoebe

Y4 · Communication

Asian/PI · neurodivergent

Open your envelope.

Imagine this is a student in one of your classes.

DIRECTIONS

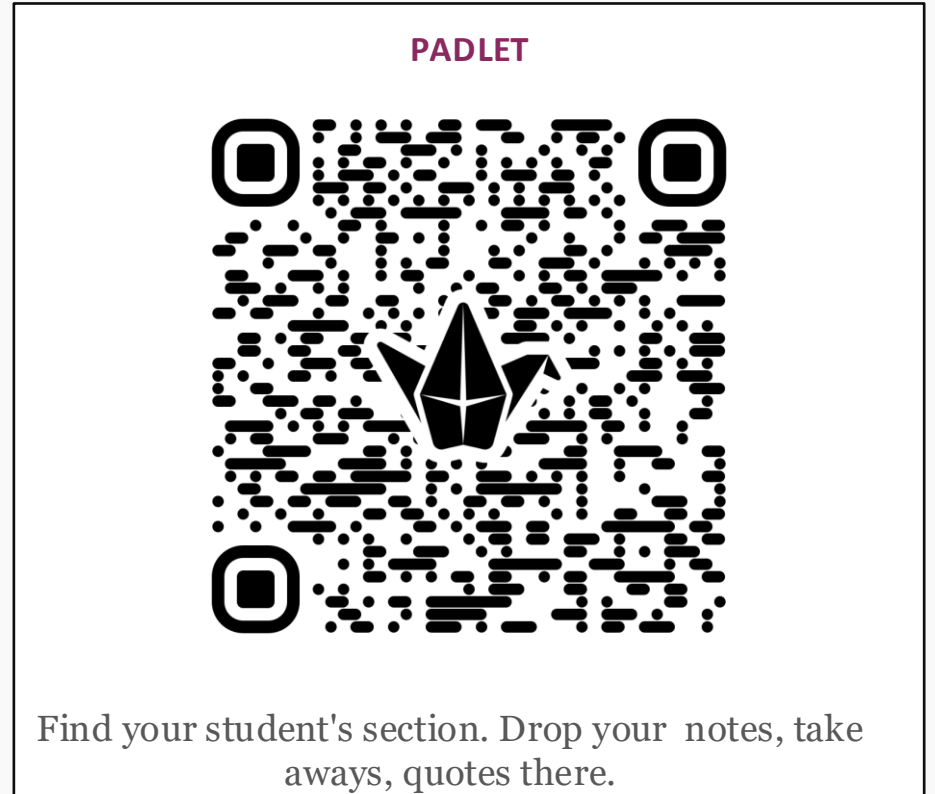
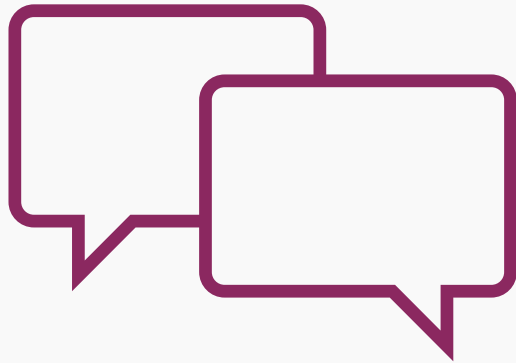
1. Read carefully. Annotate as you go.
2. Annotate what surprises you, what concerns you, what feels familiar.
3. Use the post-its to capture your reactions.
4. Don't rush. Sit with this student.



8 minutes

Now talk it through.

"What does their interview surface in your own teaching?"



6 minutes

What were we looking for?

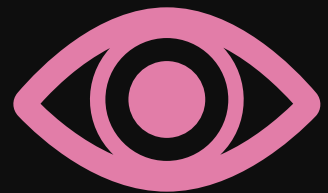
H3 · The finding that pulled me forward

Students want more opportunities to learn about AI at their institution.

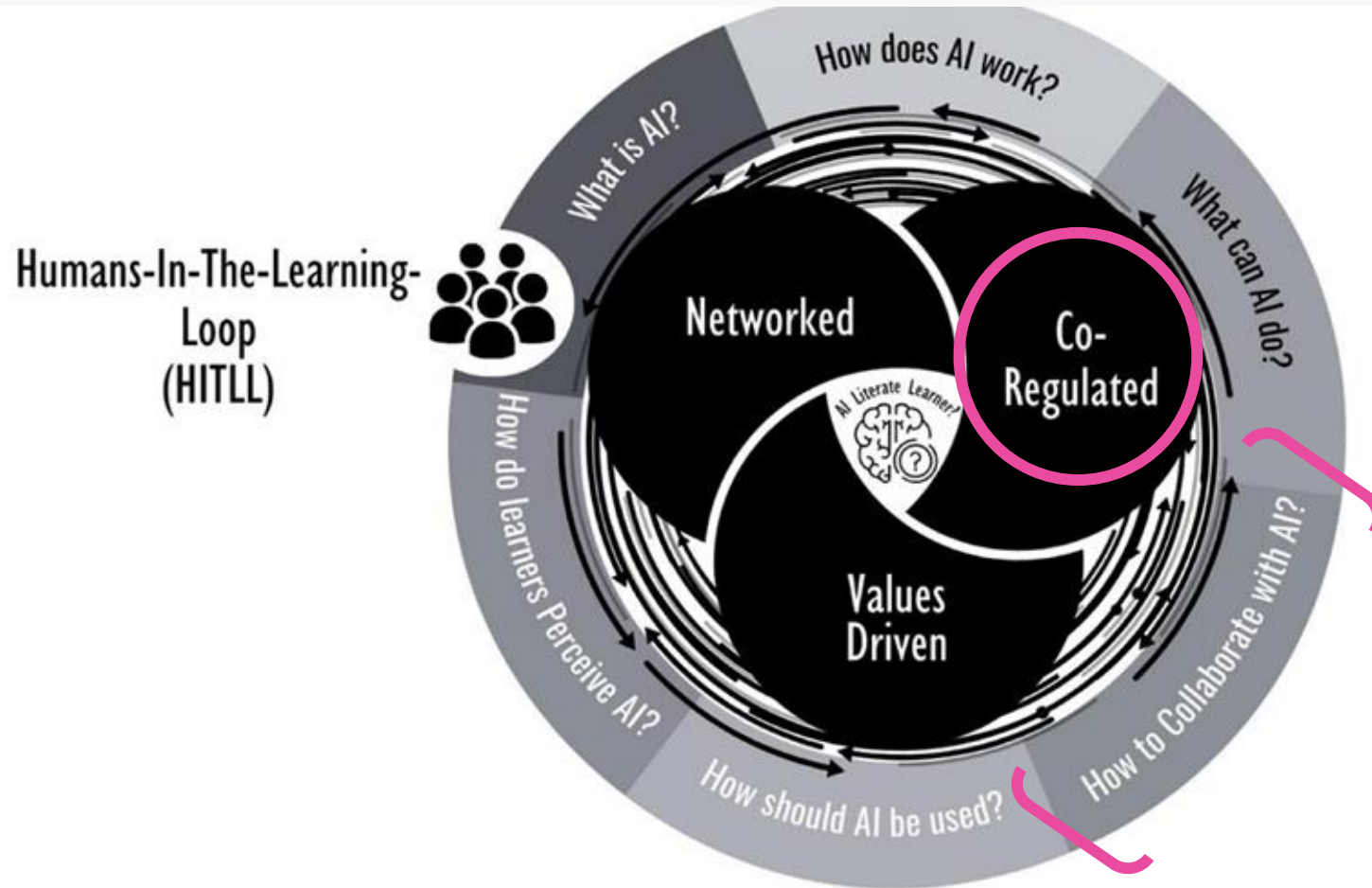
That's the thread I followed into the interviews. Here's what I saw.

02

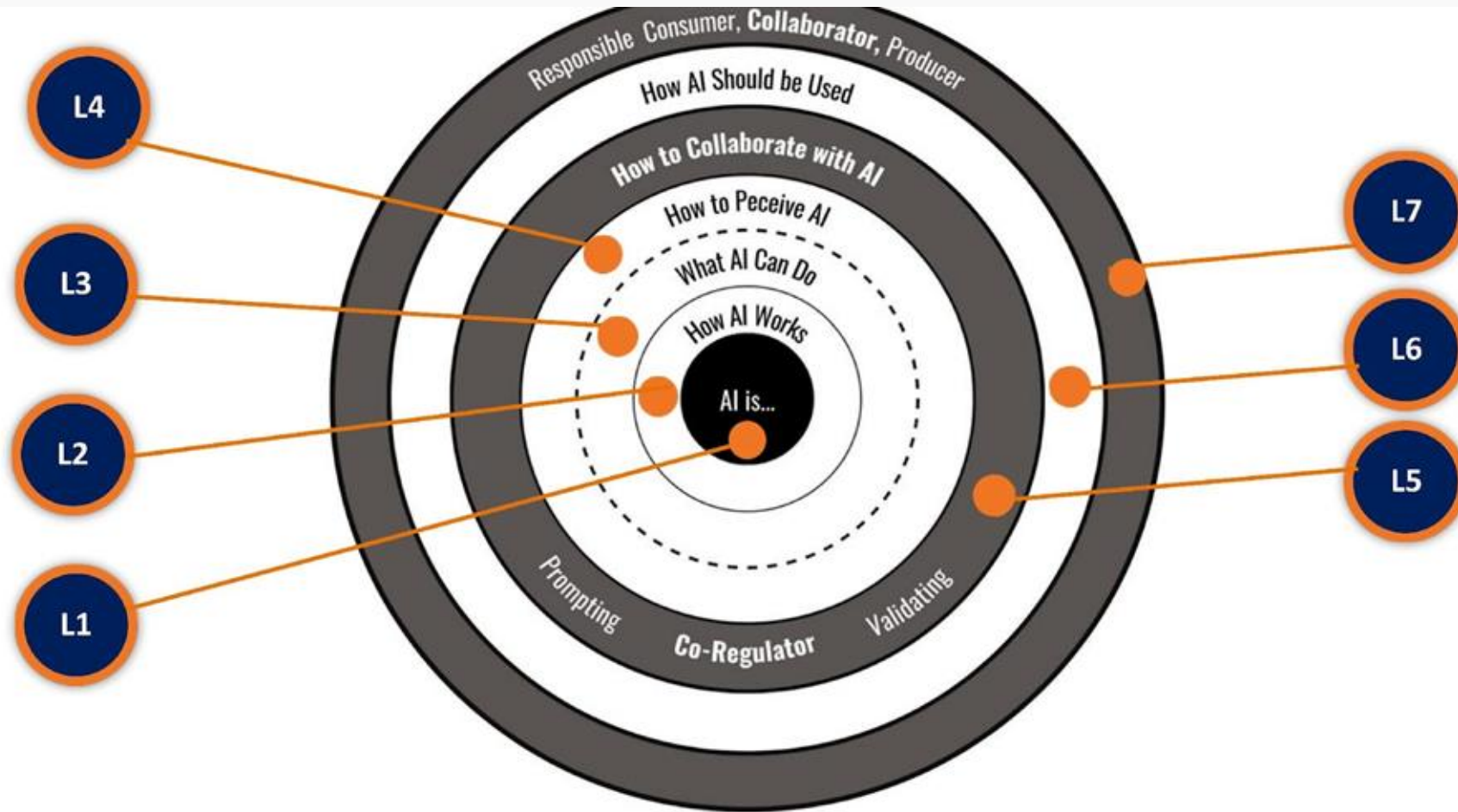
Now let me show you what I see.



What AI literacy looks like in practice.



And here's a tool you can use to foster AI-literate learners.



Alstot's Lenses for AI Literate Learning

One thing I've been working with.

Guardrail prompts: boundaries students set with AI before they ask.



THE MASTER GUARDRAIL — STUDENTS PASTE THIS FIRST

Act as a learning coach. Do not write, solve, summarize, or revise my work for me. Do not generate final answers. Instead, help me think by asking questions, identifying issues, giving feedback, and suggesting next steps.

Keeps the cognitive work where it belongs — with the learner.

Full library — discipline-specific, executive function, syllabus language — is in the gift waiting for you at the end.

03

Now what will you do.



Roll the die. Pick a question. Commit on Padlet.

4 minutes



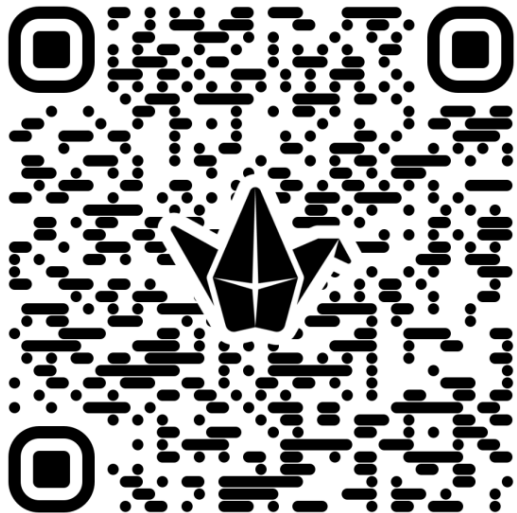
How does your current AI policy — or its absence — affect the student you encountered today?



Which lens does your policy or pedagogical approach support? How?



How could you scaffold AI integration in an assignment you typically give?



Take what you rolled or pick a different one. The die is a starting point, not a sentence.

PADLET

Same QR as before. Drop your commitment in section 10.

WORKSHEET

Use your worksheet to capture what surfaces.

References.

Alstot, M. (2026). Generative artificial intelligence and the metaliterate learner: A mixed methods study of AI-literacy and US undergraduate students' interaction within an AI-integrated knowledge economy [Doctoral dissertation, Pepperdine University].

Ludwig, D. L., & Zakrajsek, T. D. (2026). The science of learning meets AI: A practical faculty guide to purposeful integration, student engagement, and ethical practice. Routledge.